BFMC 2025 : Status Report 4
Date : Monday, March 17, 2025
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Overview of Project Status

The past month focused on the fine tuning of the car to ensure that it could perform an adequate run on the track for the qualifications. As such, much of the time was spent solving little issues that would otherwise tamper with our ability to record a run that was up to our standards. Recalibration of the speed and steering with a definite weight, recalibration of sign position estimate from the depth camera and extensive testing of the PID controller were the main subjects of our fine tuning. On top of that, the speed curve portion of the track was built and improvements brought to the GUI. This enabled us to record a satisfactory run for the qualifications.

Current Phase

Reliability is without a doubt one of (if not the) most important qualities that a self-driving car must possess. This is why calibration has been subject to such an intense study on our part. If our algorithms are able to identify the correct action to take, but the command that they send is not the one acted out, then even the most perfected algorithms are useless. Our integration testing revealed a flaw in our calibration process which will be detailed in the *Encountered Issues* section.

As was mentioned in the previous status report, relocalization from exterior reference points is essential to our car keeping the correct course. One of the reference points with which the car relocalizes are the signs, but another that we are working on improving is lane detection. Improving from last year's algorithm to make it possible to run in real time, modifications are under way to extract waypoints that can be used by the MPC planner from the lane lines on the track. The computational complexity of the polynomial fit as well as its stability are challenges that are currently being reckoned with.

User interface and user experience make a world of difference for the ones interacting with the car. Providing intuitive control, flexibility in the operations and variety in the situations make both testing and evaluating features of the car a smooth experience. Ahead of the qualifications, more work was put into the GUI to be able to properly convey the abilities of our vehicle to the judges. A refined version of the dashboard, which displays not only the speed and steering, but also the signs encountered, their distance relative to the car and their position on the map was implemented to give those evaluating a more complete picture of the qualification run.

Task Status - Overview

The team was definitely under more pressure as the qualification deadline was approaching. A week of break from classes in early March enabled us to dedicate a lot of time to preparing for qualification. Careful planning of this break week with a balance of both development and integration testing enabled us to record a run with one week to spare before the qualification deadline. A detailed status of the tasks can be found in the attached task tracker table export[1]. The main tasks will be briefly discussed in the following.

Track Design - Qualification: This task was of high priority and is complete. As this was the second time we were building the track, the process was much faster and more efficient. Careful plans needed to be

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[1] See *task_tracker_export_status_qualifications.pdf*

made for the speed curve's design, and the method of application of paint was changed. A report on the hardware detailing the process is being drafted[2].

Model Based Steering : This task was of high priority and is incomplete. Testing with both the PID and MPC yielded disappointing results : the car was unable to follow a straight line. Our hypothesis is that the system is unstable because of the two different kinematic bicycle models implemented.

Graphical User Interface : Aesthetic redesign of the icons, their positioning and the way they can be interacted with has been undertaken. In addition to that, features to display the surroundings of the car and encountered obstacles in three dimensional space was added.

Blocking Points and Encountered Issues

Jetson Corruption : A critical issue that showed up in the last week of February was the corruption of our Jetson Nano. This was critical as the Jetson is our main computing unit, and its computational capabilities cannot be replaced by those of the raspberry pi. Essentially, losing this piece of hardware meant we would not qualify. A full reinstall of the firmware, using a borrowed Jetson from the University, were required to fix our own Jetson. A detailed description of the process can be found in our documentation[3].

Voltage Regulation : During the qualifying run, we noticed that the car did not perform the same depending on the battery we were using. Further testing showed a critical flaw in our current hardware system : the powerboard we are using does not regulate the voltage, which means that the speed of the motor varies not only with respect to the input PWM but also with respect to the battery voltage. This annihilates the calibration that was carefully done. While we were able to get adequate performance for the qualification run, this issue put into question the reliability of our solution. Going forward, it will be a priority to address the voltage regulation of the input power to the motor.

Next Steps

The qualification run done, the team is determined to not take the foot off the gas. While there is no certainty, we are quite confident in our ability to qualify and are already taking measures to improve our current capabilities ahead of the finals. The qualification run enabled us to identify our weaknesses and areas that require improvement. Given the extended time period between qualifications and finals, the nature of the tasks that to be undertaken can be more radical.

The most critical issue at hand is the regulation of the voltage provided to the motor. Xin Ya is currently investigating the issue. The new powerboards we designed don't do voltage regulation *a priori*, but we have found a voltage regulator in our spare parts that we will attempt to use. Extensive testing and characterization of the sensor under different conditions will be required.

The addition of a rotary encoder to the driving shaft of the car is also being considered and will be attempted. This will give us external feedback for the speed of our vehicle, which is essential so that we don't have to only rely on the PWM calibration of the motor. This should provide a backup to the calibration that was done and enable data fusion should such analysis be necessary.

Lane detection is another focal point of the coming month. Last year's algorithm showed promise but could not be implemented in real time. A great deal of time will be spent refining and improving the algorithm such that meaningful and reliable waypoints can be extracted from the lanes.

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[2] See *pothole_avoidance_hardware_report.pdf*
[3] See *suivi_jetson.pdf*